U-Pb – Oxygen spot matcher user guide

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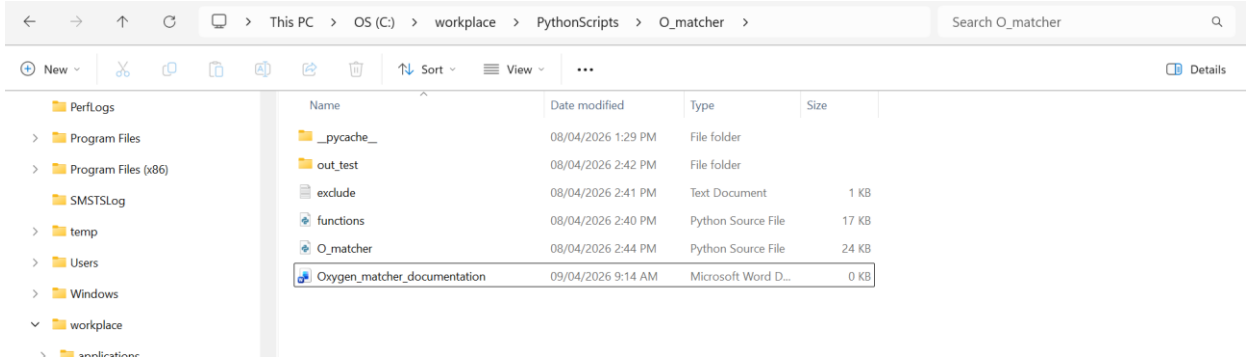
Directory structure and setup

The workflow relies on two separate .py (Python) scripts: functions.py and O_macher.py (Fig. 1). The scripts need to be located within the same directory. Python scripts need to be located inside “C:\workplace” and the creation of a sub-directory within “C:\workplace” to store these files is recommended (Fig. 1).



The functions.py is an auxiliary script that is called by O_matcher.py, the main script, to process the data.

The O_matcher.py script is called from the Anaconda prompt. It matches the U-Pb

Figure 1: Directory structure.

analyses to their corresponding Oxygen isotopic analyses. It gives the user the option to export three different levels of processed data (see **Workflow** section).

Important: the script relies on the Python package xlrd, which is not installed by default when installing Anaconda. The rest of the packages are installed by default.

Running the script

If Python was installed using Anaconda there will be a software called Anaconda Prompt available on your computer. Running the script from Anaconda Prompt is recommended to avoid having to set up path variables. Once Anaconda Prompt is open navigate to the script location using the “cd” (change directory) command and typing the directory path on the prompt (Fig. 2). Once in the correct directory run the script by typing “python O_Matcher.py” and pressing enter.

```
Anaconda Prompt

(base) C:\Users\MIPSAMP>cd C:\workplace\PythonScripts\O_matcher

(base) C:\workplace\PythonScripts\O_matcher>python O_matcher.py
```

Figure 2: Anaconda prompt interface. First step is navigating to the correct directory. In this case the script was located in "C:\workplace\PythonScripts\O_Matcher", but it would work in any directory inside "C:\workplace". After confirming the directory, run the script by typing python O_matcher.py

Workflow

Once the script is running, it will prompt the user to input a name for the output directory where the generated tables will be exported to.

The required inputs are:

1. U-Pb file directory – this one is a path containing the .xlsx, .xls or .txt files with the geochron data. Files should be in the standard GSWA format. Currently only SHRIMP formatted files are working because the header expects the ^{204}Pb-corrected data

2. Oxygen file directory – path containing the RAW oxygen data files (e.g. V:\GS54_GeochronologyLaboratory\Projects\Isotope_studies\ES46_2024-25\Oxygen\3_data_received\Batch_19\1_Raw_Data_Files). The script expects the "3-CorrectedData" sheet.

Then, the user will be prompted to select which table to generate:

- Option 1: Export intermediate-level table (unformatted) to check whether points are primary or secondary and a WAGIMS-formatted reference material spot-level analyses.

- Option 2: Export a WAGIMS-formatted table with spot-level information

- Option 3: Export a sample-level table (unformatted).

The option will then be prompted to input the path to the directory containing the U-Pb files and then for the directory containing the Oxygen files, these prompts happen regardless of the selected option. The script will flag any U-Pb files that do not conform

```
(base) C:\workplace\PythonScripts\O_matcher>python O_matcher.py
Output directory:out_test
Select file to generate:
1. Intermediate processing file
2. Final WAGIMS spot-level output
3. Geoview sample-level output
Your option: 1
U-Pb files directory: V:\GS54_GeochronologyLaboratory\Projects\Isotope_studies\ES46_2024-25\Oxygen\2_sent\Batch_19\Batch_19_U-PbTables
Oxygen files directory: V:\GS54_GeochronologyLaboratory\Projects\Isotope_studies\ES46_2024-25\Oxygen\3_data_received\Batch_19\1_Raw_Data_Files
Check 242652.1.txt manually
C-O-OH-GSWA-G11-29.xlsx
C-O-OH-GSWA-G12-13.xlsx
C-O-OH-GSWA-G12-15.xlsx
C-O-OH-GSWA-G15-10.xlsx
C-O-OH-GSWA-G15-22.xlsx
C-O-OH-GSWA-G18-18.xlsx
C-O-OH-GSWA-G20-3.xlsx
C-O-OH-GSWA-G21-1.xlsx
C-O-OH-GSWA-G21-10.xlsx
C-O-OH-GSWA-G21-22.xlsx
C-O-OH-GSWA-G21-25.xlsx
C-O-OH-GSWA-G21-26.xlsx
C-O-OH-GSWA-G22-16.xlsx
C-O-OH-GSWA-G22-27.xlsx
C-O-OH-GSWA-G22-30.xlsx
C-O-OH-GSWA-G23-12.xlsx
```

Figure 3: Example run of the script.

to GSWA standard formatting to be checked manually, and it will print out the found oxygen files it found (Fig. 3). The script will also flag the samples that didn't contain any matched spots for manual investigation.

For any option selected, the user is prompted to select a name for the output files. The extension (.xlsx) **must** be included.

The intermediate table exported by option 1 is for data processing. For ease of comparison between standards and unknowns from the same session, a "file" column is also generated. The user should also look for unknowns that have oxygen information but no U-Pb information, as this indicates that no match has been found, which is usually a result of different spot name formatting (e.g. U-Pb spot being 6-1 and Oxygen spot being 06-1). In this case, the spot name should be edited in the original table, and the script should be run again.

Option 3 reads in the "exclude.txt" file, that contains the spot names, in this case the oxygen analysis ID (e.g. O-GSWA-G22-16-251159@36-1). This is to exclude spots that have a secondary signature from the sample-level calculations. The final sample-level table summarises the information for the $\delta^{18}O$ values, and contains the following columns: median, propagated 1 sigma uncertainty, average, standard deviation, weighted mean, confidence interval on the weighted mean, MSWD and confidence level. By default, all confidence intervals are calculated at the 95% confidence level. Exclusion of outliers is done by excluding spots that are more than 2.5σ away from the mean.